3 Problems

The problems in this section were mainly taken from [4] and [2].

Problem 1 (2002 Hong Kong TST)**.** For an irrational number x, let x' be the integer closes to x. Define $\langle x\rangle = |x - x'|$. Show that for every irrational number y, the minimum of the numbers $\langle y\rangle, \langle 2y\rangle, \ldots, \langle 2001y\rangle$ is less than $\frac{1}{2001}$.

Problem 2 (2000 Putnam A2)**.** Prove that there are infinitely many triples of consecutive integers each of which is a sum of two squares.

Problem 3. Find all triangles whose sides are consecutive integers and areas are also integers.

Problem 4. Determine all pairs (k, m) such that $k < m$ and $1 + 2 + \cdots + k = (k + 1) + (k + 2) + \cdots + m$.

Problem 5. Find all triangular numbers that are perfect squares.

Problem 6. Prove that there are infinitely many positive integers n such that $n^2 + 1$ divides $n!$.

Problem 7. Show that there are infinitely many integers n such that $2n + 1$ and $3n + 1$ are perfect squares, and that such n must be multiples of 40.

Problem 8. Prove that if n is a positive integer such that both $3n + 1$ and $4n + 1$ are perfect squares, then n is divisible by 56.

Problem 9 (1986 USAMO)**.** The root-mean-square of the numbers $a_1, a_2, \ldots, a_n$ is defined as $\sqrt{\dfrac{a_1^2 + a_2^2 + \ldots + a_n^2}{n}}$. What is the smallest integer n, greater than 1, for which the root-mean-square of the first n positive integers is an integer?

Problem 10. For the sequence $a_n = \left\lfloor \sqrt{n^2 + (n+1)^2} \right\rfloor$, prove that there are infinitely many n's such that $a_n - a_{n-1} > 1$ and $a_{n+1} - a_n = 1$.

Problem 11. Prove that the only positive integer solution of $5^a - 3^b = 2$ is $a = b = 1$.

Problem 12. Show that the equation $a^2 + b^3 = c^4$ has infinitely many integer solutions.

Problem 13 (1999 Bulgaria)**.** Prove that the equation $x^3 + y^3 + z^3 + t^3 = 1999$ has infinitely many integer solutions.

Problem 14 (1995 IMO Proposal by USA Leader Titu Andreescu)**.** Find the smallest positive integer n such that $19n + 1$ and $95n + 1$ are both integer squares.

Problem 15 (2003 Chinese Team Training)**.** Let $x_0 + y_0\sqrt{2003}$ be the fundamental solution of $x^2 - 2003y^2 = 1$. Find all positive solutions (x, y) such that each of the prime factors of x divides x_0.

Problem 16 (2002 IMO Shortlist N4)**.** Is there a positive integer m such that the equation

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{abc} = \frac{m}{a + b + c}$$

has infinitely many solutions in positive integers a, b, c?

Problem 17 (Evan Chen)**.** A *fissile square* is a positive integer which is a perfect square, and whose digits form two perfect squares in a row, which will be called the *left square* and the *right square.* For example, 49, 1444, and 1681 are fissile squares (decomposing as $4 \mid 9$, $144, \mid 4$, and $16 \mid 81$ respectively). Neither the left nor the right square may begin with the digit 0. Which perfect squares are the right square of infinitely many fissile squares?

Problem 18 (2005 IMO Shortlist N5)**.** Prove that there exist infinitely many positive integers n such that $p = nr$, where p and r are respectively the semiperimeter and the inradius of a traingle with integer side lengths.

Problem 19 (2019 IMO Shortlist N8)**.** Let a and b be two positive integers. Prove that the integer

$$a^2 + \left\lceil \frac{4a^2}{b} \right\rceil$$

is not a square.

Problem 20 (2019 US Math Competition Association Premier P8)**.** Find all pairs of positive integers (m, n) such that $(2^m - 1)(2^n - 1)$ is a perfect square.